

Comparison of Classical and Hybrid Quantum Models for Credit Card Fraud Detection

Motivation

Financial fraud is a major problem in modern digital economies. It refers to intentional deception in financial transactions in order to gain money, assets, or services illegally. Fraudulent activities affect banks, financial institutions, and customers, causing billions of dollars in losses every year. Apart from financial loss, fraud also reduces trust in digital payment systems and online banking platforms.

To reduce these risks, financial institutions use **fraud detection systems** that analyse transaction data and automatically identify suspicious behaviour. In machine learning terms, fraud detection is typically modelled as a **binary classification problem**, where transactions are classified as either normal or fraudulent.

0 → Normal Transaction
1 → Fraudulent Transaction

However, fraud detection presents several challenges. One of the main issues is **extreme class imbalance**. Fraudulent transactions are very rare compared to legitimate transactions. In the dataset used for this study, there are **284,807 total transactions but only 492 fraud cases**, which corresponds to approximately **0.17% fraud**. Because of this imbalance, traditional metrics like accuracy can be misleading, since a model may achieve very high accuracy while still failing to detect fraud effectively.

Another challenge is that fraudulent patterns are often **complex and hidden within large volumes of transaction data**. Fraudsters continuously adapt their strategies, making it difficult for traditional machine learning models to capture subtle relationships between features.

Quantum computing offers a promising direction for addressing such complex problems. Quantum systems operate using principles such as **superposition and entanglement**, allowing information to be represented in a **high-dimensional Hilbert space**. This enables quantum models to represent complex relationships between features that may be difficult for classical models to learn.

In hybrid quantum-classical models, **quantum circuits can be used as nonlinear feature transformation layers** within a neural network. These quantum layers can potentially capture higher-order feature interactions and enhance the expressive power of the model. As a result, hybrid quantum-classical neural networks may improve the ability to detect subtle fraud patterns in financial transaction data.

Therefore, this study explores whether integrating quantum circuits with classical neural networks can improve the performance of fraud detection systems.

Methodology

The objective of this study is to develop and evaluate a **hybrid quantum-classical neural network model** for detecting fraudulent financial transactions.

Dataset

The experiment uses the **Credit Card Fraud Detection Dataset** available on Kaggle. This dataset contains **284,807 credit card transactions**, among which **492 transactions are labelled as fraud**, representing approximately **0.17% of the total data**.

The dataset contains the following features:

- **V1–V28**: PCA-transformed numerical features (used to protect confidentiality of original variables)
- **Time**: Time elapsed between transactions
- **Amount**: Transaction amount
- **Class**: Target variable indicating fraud (1) or normal transaction (0)

Because the dataset is highly imbalanced, special attention is required when training and evaluating models.

Data Preprocessing

Before training the model, several preprocessing steps are performed:

1. **Data normalization** is applied to numerical features such as the transaction amount to ensure consistent scaling.
2. The dataset is **split into training and testing sets** to evaluate the model's performance on unseen data.
3. Due to the class imbalance, evaluation metrics beyond accuracy, such as **precision, recall, and F1-score**, are considered.

Hybrid Quantum-Classical Model

The proposed model combines classical neural network components with a **quantum circuit layer**. The classical part processes the input features and prepares them for the quantum layer.

The quantum component performs the following steps:

1. **Encoding classical data into quantum states** using parameterized quantum gates.

- 2. Applying **variational quantum circuits** to transform the encoded data.
- 3. Measuring the quantum states to generate outputs that are fed back into the classical neural network.

This hybrid approach allows the model to leverage both **classical computation efficiency** and the **expressive representation power of quantum circuits**.

Model Training and Evaluation

The model is trained using labelled transaction data where the objective is to minimize classification error between predicted and actual labels. During training, the parameters of both classical layers and quantum circuits are optimized.

The trained model is then evaluated on the test dataset using metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide a better understanding of how effectively the model detects fraudulent transactions while minimizing false positives.

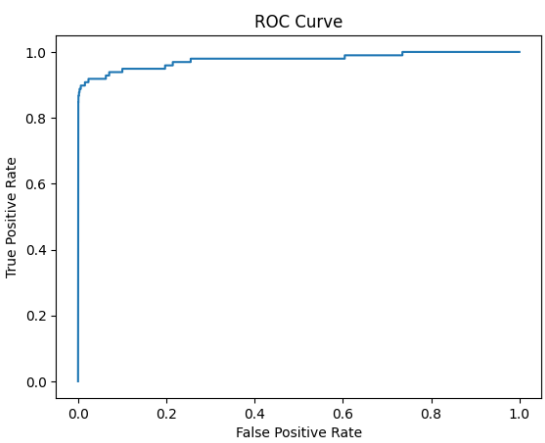
Results

Classical Model Results:

Class	Precision	Recall	F1-score	Support
0.0	1.00	0.99	1.00	56,864
1.0	0.20	0.90	0.32	98

Metric	Value
Accuracy	0.99
Macro Avg Precision	0.60
Macro Avg Recall	0.95
Macro Avg F1-score	0.66
Weighted Avg Precision	1.00
Weighted Avg Recall	0.99
Weighted Avg F1-score	1.00
ROC-AUC	0.9764

Training Time	56.15 s
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Quantum Model Results:

Class	Precision	Recall	F1-score	Support
0.0	1.00	0.96	0.98	56,864
1.0	0.02	0.39	0.03	98

Metric	Value
Accuracy	0.96
Macro Avg Precision	0.51
Macro Avg Recall	0.67
Macro Avg F1-score	0.50
Weighted Avg Precision	1.00
Weighted Avg Recall	0.96
Weighted Avg F1-score	0.98
ROC-AUC	0.8166

Training Time	1076.85 s
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